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Characterisation of semiconducting V_2O_5 – Bi_2O_3 – TeO_2 glasses through ultrasonic measurements

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Abstract

Tellurite containing vanadate $(50 - x)V_2O_5$ – xBi_2O_3 – $50TeO_2$ glasses with different bismuth ($x = 0, 5, 10, 15, 20$ and 25 wt%) contents have been prepared by rapid quenching method. Ultrasonic velocities (both longitudinal and shear) and attenuation (for longitudinal waves only) measurements have been made using a transducer operated at the fundamental frequency of 5 MHz in the temperature range from 150 to 480 K. The elastic moduli, Debye temperature, and Poisson's ratio have been obtained both as a function of temperature and Bi_2O_3 content. The room temperature study on ultrasonic velocities, attenuation, elastic moduli, Poisson's ratio, Debye temperature and glass transition temperature show the absence of any anomalies with addition of Bi_2O_3 content. The observed results confirm that the addition of Bi_2O_3 modifier changes the rigid formula character of TeO_2 to a matrix of regular TeO_3 and ionic behaviour bonds (NBOs). A monotonic decrease in velocities and elastic moduli, and an increase in attenuation and acoustic loss as a function of temperature in all the glass samples reveal the loose packing structure, which is attributed to the instability of TeO_4 trigonal bipyramid units in the network as temperature increases. It is also inferred that the glasses with low Bi_2O_3 content are more stable than with high Bi_2O_3 content.

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1. Introduction

Semiconducting oxide glasses containing transition metal ions are more important due to their increasing applications [1] in different fields. Sim-

ilarly, tellurite glasses, a distinct type of non-crystalline materials, having unique structure and properties are of special interest in view of their applications [2] in IR domes, laser windows and multifunctional optical components. Tellurium oxide (TeO_2) is a conditional glass former [3] and forms glass only with a modifier such as alkali, alkaline earth and transitional metal oxides (TMO) or other glass formers. In a binary tellurite glasses, the basic structural unit of TeO_4 is trigonal bipyramid (tbp) with lone pair of electrons and the

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structural units take the Te–O–Te bond for glass formation [4].

The structure and physical properties of non-crystalline tellurite glasses with semiconducting oxides has attracted many researchers [5,6]. This is mainly due to the numerous applications of these glasses in electronics and other industries, and also to the fundamental interest in understanding the microscopic mechanism involved. For the same amount of charge carrier, vanadium tellurite glasses are highly conductive than the vanadate phosphate glasses or other transitional metal oxide glasses [7]. The semiconducting properties of vanadium tellurite glasses containing oxides like SnO, ZnO and Sb₂O₃ has been analysed through conductivity measurements [8–10]. The change in structure and physical properties of binary and ternary tellurite glasses such as V₂O₅–TeO₂, Li₂O–TeO₂, TeO₂–BaO–TiO₂ and V₂O₅–ZnCl₂–TeO₂ have been investigated using ESR spectra [11], thermal studies [12], Raman spectra [13] and IR spectra [14]. The electrical conductivity and transport behaviour of vanadium tellurite glasses with the addition of iron oxide [15] and lead fluoride [16] have been reported extensively.

Ultrasonic non-destructive characterisation of materials is a versatile tool for investigating the change in microstructure, deformation process and mechanical properties of materials over a wide range of temperatures [17]. This is possible due to the close association of the ultrasonic waves with elastic and inelastic properties of the materials. It is also due to the availability of different frequency range and many modes of vibration of the ultrasonic waves to probe into the macro, micro and submicroscopic levels. Recent ultrasonic studies on BaTiO₃ doped vanadate lead glasses [18], bismuth lead tellurite [19], vanadium tellurite [20] and copper tellurite glasses [21] show interesting features with addition of second/third component. The structural stability of the glasses with change in temperature is also an interesting observation made in the above studies.

In view of the electronic and industrial application of vanadium tellurite glasses, the availability of the structural changes, structural stability and elastic properties over a wide range of temperature will suit the requirements. Although, the

glass forming region and electrical conductivity of vanadate bismuth [22] glasses (V₂O₅–Bi₂O₃–TeO₂) with addition of TeO₂ have been studied well, while the ultrasonic non-destructive characterisation of this glasses both at room temperature and the effect of temperature has been found to be not reported. Therefore, in the present study, we have made an attempt to explore the structural changes, structural stability and physical properties of vanadium tellurite glasses with different Bi₂O₃ contents using ultrasonic measurements. We have measured the ultrasonic velocities, elastic moduli, attenuation (longitudinal waves only), Poisson's ratio and Debye temperature as a function of temperature from 150 to 480 K in vanadium tellurite glasses with different Bi₂O₃ contents.

2. Experimental

2.1. Preparation of glasses

Tellurite containing vanadium (50 – *x*)V₂O₅–*x*Bi₂O₃–50TeO₂ oxide glasses with different bismuth contents *x* = 0, 5, 10, 15, 20 and 25 wt% (hereafter, termed as VBT0, VBT5, VBT10, VBT15, VBT20 and VBT25, respectively) have been prepared by rapid quenching method. The analytical grade materials of purity more than 99.9% of V₂O₅, Bi₂O₃ and TeO₂ chemicals were used to prepare glass samples. The required amount (approximately 10 g) in wt% of chemicals in powder form was weighed using a digital balance having an accuracy of ±0.1 mg. The homogeneity of the mixture of chemicals was achieved by repeated grinding using an agate mortar. The mixture was preheated in atmospheric condition at 523 K for 5 h in order to remove the H₂O and CO₂. The preheated mixture was again ground for 2 h to reduce the particle size. The mixture was then melted using a high-grade alumina crucible in a furnace for about 1.30 h, till bubble free liquid was obtained. The melting temperature varies from 1023 to 1080 K depending on the Bi₂O₃ content. The mixture was stirred intermediately in order to obtain homogeneous mixture. The molten mixture was poured into a copper block. It was rapidly pressed using another thick copper block.

This results in the formation of VBT glass samples. These glass samples were prepared in the form of circular discs of 1 cm diameter and 0.3–0.5 cm thickness for velocity and attenuation measurements. A similar procedure was employed to prepare the VBT0, VBT5, VBT10, VBT15, VBT20 and VBT25 glass samples.

In order to achieve proper impedance matching between the sample and the transducer for the propagation of the ultrasonic waves into the glass samples, the opposite surfaces of the glass sample were polished using fine lapping paper. The polished samples were etched with 1% of HCl for a few minutes. The plane parallelism between the opposite faces of the glass sample was checked using a surface plate and a dial gauge. The variation in the sample thickness is found to be $\pm 5 \mu\text{m}$.

2.2. Ultrasonic velocity measurements

Ultrasonic velocity measurements both at low (liquid nitrogen to 303 K) and high temperature (303–480 K) have been made in all the glass samples using experimental set ups in the authors' laboratory [23]. The block diagram of the experimental set up used for velocity and attenuation measurements is shown in Fig. 1.

A high power ultrasonic process control system and a digital storage oscilloscope were employed for ultrasonic rf signal recording. Precise transit time measurement were carried out employing

cross-correlation technique for both longitudinal and transverse waves using a transducer operating at the fundamental frequency of 5 MHz. The precise ultrasonic transit time measurements were made employing various steps involved as followed in our earlier studies [24]. Using the above procedure, one can measure the transit time with an accuracy of $\pm 0.2 \text{ ns}$ resolution. The other phenomenon to be consider in the determination of accurate velocity in the sample is the variation in the thickness of the glass sample and plane parallelism between the opposite faces. As indicated earlier these are minimised by surface grinding of the glass sample and obtaining the plane parallel sample with an accuracy of $\pm 5 \mu\text{m}$. In our measurements, the over all variation in velocity measurements is $\pm 5 \text{ m s}^{-1}$ which corresponds to a percentage change in velocity of about 0.1%.

2.3. Attenuation measurements

The ultrasonic attenuation (α) in all the glass samples were measured (only for longitudinal waves) by measuring the peak amplitude of successive back wall echoes of ultrasonic signals from the glass sample using the relation

$$\alpha = \left(\frac{-20}{2(m-n)d} \right) \log \left(\frac{I_m}{I_n} \right), \quad (1)$$

where I_m and I_n are, respectively, the maximum amplitude (voltage) of the m th and n th pulse

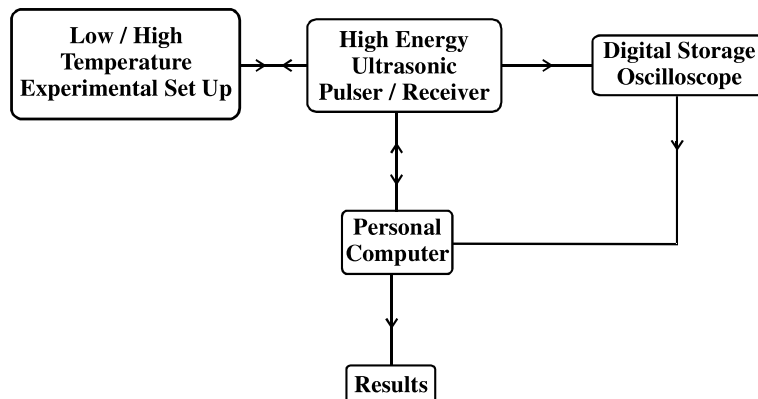


Fig. 1. Block diagram of the experimental set up.

echoes. A constant pressure was maintained between the transducer and glass sample during the contact measurement. Necessary care was taken to avoid the near field effects and also in the selection of suitable couplant. Among the various available couplants, the machine oil and non-aqueous stopcock grease respectively for high and low temperature measurements were found to be more suitable to get steady back wall echo train in the oscilloscope screen. Thus, the percentage error in the attenuation measurements is $\pm 2\%$.

2.4. Density measurements

Density of all glass samples was measured employing Archimedes principle with carbon tetrachloride as buoyant using the relation

$$\rho = \left(\frac{W_a}{W_a - W_b} \right) \rho_b, \quad (2)$$

where W_a is the glass sample weight in air, W_b the glass sample weight in buoyant and ρ_b the density of the buoyant. All the weight measurements have been made using a digital balance having an accuracy of ± 0.1 mg. The experiment was repeated for five times to get the accurate value. The overall accuracy in the density measurement is ± 0.5 mg and hence, the percentage error in the measurement of density is $\pm 0.05\%$.

2.5. X-ray diffraction and thermogravimetric measurements

X-ray diffraction (XRD) analysis and scanning electron microscope (SEM) studies in all the glass samples have been made to confirm the amorphous nature. The thermal behaviour of all glass samples has been studied from differential thermal analysis (DTA) and thermogravimetric (TG) measurements. The TG-DTA of all glass samples were made with a heating rate of $10^\circ\text{C}/\text{min}$. The glass transition temperature (T_g) values are determined from the TG-DTA curve. The accuracy in the measurement of T_g is ± 2 K. In addition, the elements present in the glass samples with different Bi_2O_3 contents were evaluated by energy dispersive

analysis X-rays (EDAX). The estimated composition present in different glass samples was found to be slightly lesser than the corresponding nominal elemental composition values (before glass formation). The difference in the composition is due to the evaporation losses, uncertainty in the chemical analysis, and the presence of impurity phases formed, which we have not taken into account.

3. Determination of elastic constants

Elastic (longitudinal, shear, bulk and Young's) moduli, Debye temperature and Poisson's ratio of $(50-x)\text{V}_2\text{O}_5-x\text{Bi}_2\text{O}_3-50\text{TeO}_2$ glasses with different Bi_2O_3 contents have been determined from the measured ultrasonic velocities and density using the standard relations [18]:

Bulk modulus

$$K = L - (4/3)G; \quad (3)$$

Young's modulus

$$E = (1 + \sigma)2G, \quad (4)$$

where L is the longitudinal modulus ($L = U_L^2 \rho$) and G the Shear modulus ($G = U_S^2 \rho$). σ is the Poisson's ratio and is given by

$$\sigma = \left(\frac{(L - 2G)}{2(L - G)} \right); \quad (5)$$

Debye temperature [18] of the glass samples was obtained from the relation

$$\theta_D = \left(\frac{h}{K_B} \right) \left(\frac{3PN_A}{4\pi V_a} \right)^{1/3} U_m, \quad (6)$$

where h is the Planck's constant, K_B the Boltzmann's constant, N_A the Avogadro number, V_a the molar atomic volume calculated from the effective molecular weight and the density (i.e. M/ρ), P the number of atoms in the molecular formula and U_m the mean sound velocity defined by the relation

$$U_m = \left(\frac{1}{3} \left(\frac{2}{U_S^3} + \frac{1}{U_L^3} \right) \right)^{-1/3}. \quad (7)$$

The relative percentage change in velocity was obtained employing the relation

$$\left(\frac{\Delta U}{U}\right) = \left(\frac{U - U_{\min}}{U_{\min}}\right), \quad (8)$$

where $U_{\min}$ is the minimum sound velocity in the experimental temperature range from 150 to 480 K and U the sound velocity (longitudinal or shear) at temperature T K.

The V-ion density (N) of the glass sample was calculated using the relation

$$N = \left[\left(\frac{\rho \text{ wt}\%(\text{V}_2\text{O}_5)}{\text{Mwt}(\text{V}_2\text{O}_5)} \right) N_A \right], \quad (9)$$

where $\text{wt}\%(\text{V}_2\text{O}_5)$ is the weight fraction of V_2O_5 and $\text{Mwt}(\text{V}_2\text{O}_5)$ the molecular weight of V_2O_5 .

4. Results

4.1. X-ray diffraction and scanning electron microscopy

XRD pattern of VBT0, VBT5, VBT10, VBT15, VBT20 and VBT25 glass samples show the absence of any characteristic peaks confirms the glassy nature in all the glass samples. A typical XRD pattern for VBT5 and VBT25 glass samples are shown in Fig. 2. The absence of any microstructure on scanning electron micrograph in all the glass samples confirms the glassy nature. Fig. 3

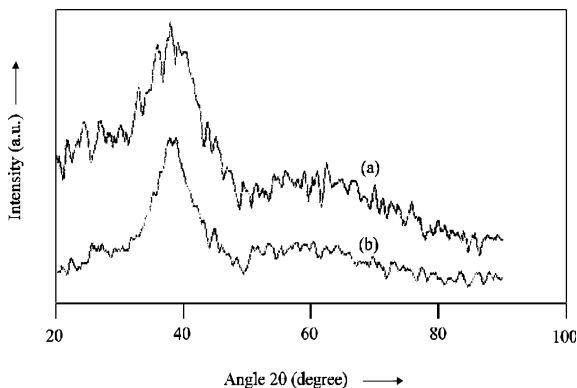


Fig. 2. XRD pattern with $\text{CuK}\alpha$ radiation of two typical glasses: (a) VBT5 and (b) VBT25 showing amorphous character.

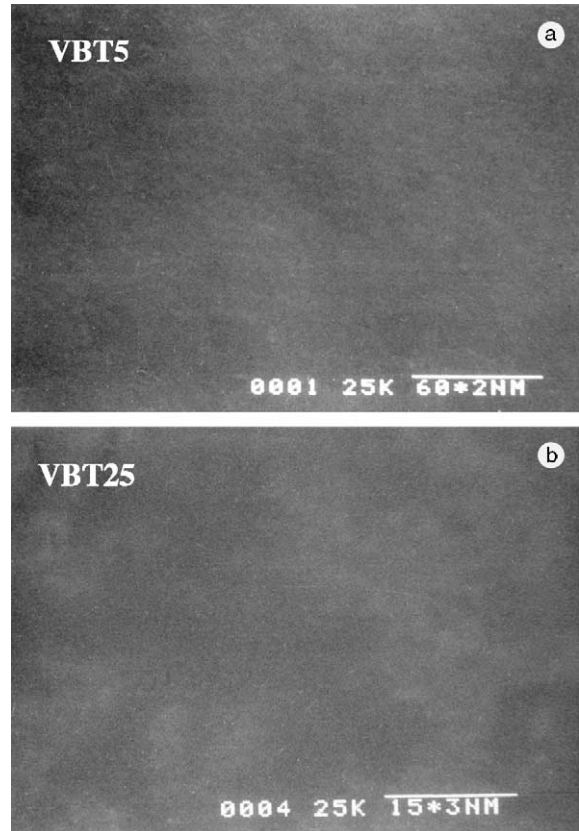


Fig. 3. Scanning electron micrographs of two typical glasses: (a) VBT5 and (b) VBT25 showing amorphous character.

shows the SEM micrograph for a typical VBT5 and VBT25 glass samples.

4.2. Thermal analysis

DTA and TG curves for VBT0, VBT5, VBT10, VBT15, VBT20 and VBT25 glass samples with different Bi_2O_3 contents have been obtained. The T_g values in all glass samples were determined from the observed endothermic dip in the TG-DTA curve. The composition dependence of T_g in the present glass system is shown in Table 1. The glass transition temperature (T_g) increases with increase in Bi_2O_3 content without showing any anomalous behaviour as shown in Table 1. The dependence of T_g with change in molar volume is given in Fig. 4.

Table 1

Glass composition, density, ultrasonic velocities, attenuation, ion density and glass transition temperature, along with the sample name of $(50 - x)\text{V}_2\text{O}_5 - x\text{Bi}_2\text{O}_3 - 50\text{TeO}_2$ glasses

Sample name	Nominal compositions			Density, ρ (kg m^{-3})	Ultrasonic velocity		Attenuation, α (dB cm^{-1})	Ion density, $N \times 10^{28}$ (m^{-3})	Transition temperature, T_g^a ($^{\circ}\text{C}$)	Molar atomic volume, V_a ($\text{cm}^3 \text{mol}^{-1}$)
	V_2O_5 (wt%)	Bi_2O_3 (wt%)	TeO_2 (wt%)		U_L (m s^{-1})	U_S (m s^{-1})				
VBT0	50	0	50	3995.6	3655	2096	0.973	1.41	230.8	42.73
VBT5	45	5	50	4376.4	3591	2056	1.187	1.28	254.6	42.26
VBT10	40	10	50	4796.9	3507	1995	1.032	1.16	272.9	41.52
VBT15	35	15	50	5188.4	3416	1937	0.929	1.03	293.3	41.12
VBT20	30	20	50	5624.0	3330	1882	0.800	0.88	309.3	40.46
VBT25	25	25	50	6031.4	3250	1833	1.030	0.75	395.6	40.08

^a Error in the measurement of T_g is ± 2 K.

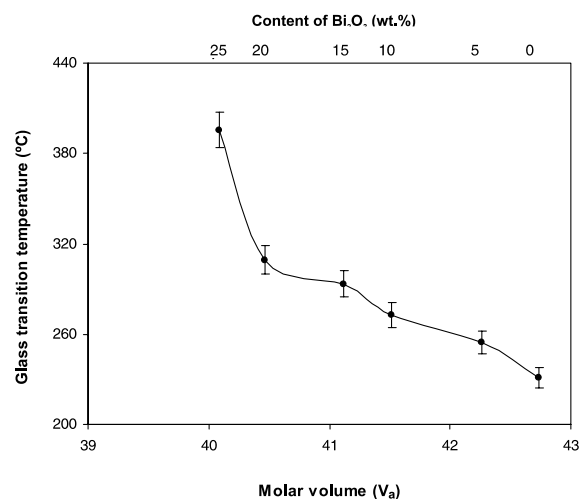


Fig. 4. Plot of molar volume with change in glass transition temperature. The lines are drawn for guides to the eye.

4.3. Ultrasonic studies

The experimental values of density, ultrasonic velocity both longitudinal and shear, and attenuation of the tellurite containing vanadate oxide glasses for different Bi_2O_3 contents $x = 0, 5, 10, 15, 20$ and 25 wt% at room temperature along with the V-ion density and glass composition are given in Table 1. The Bi_2O_3 dependence of density and attenuation show a continuous increase as shown in Fig. 5(a). The variation of ion density (N) with Bi_2O_3 content shows a continuous decrease as shown in Fig. 5(b). For a fixed temperature, say at

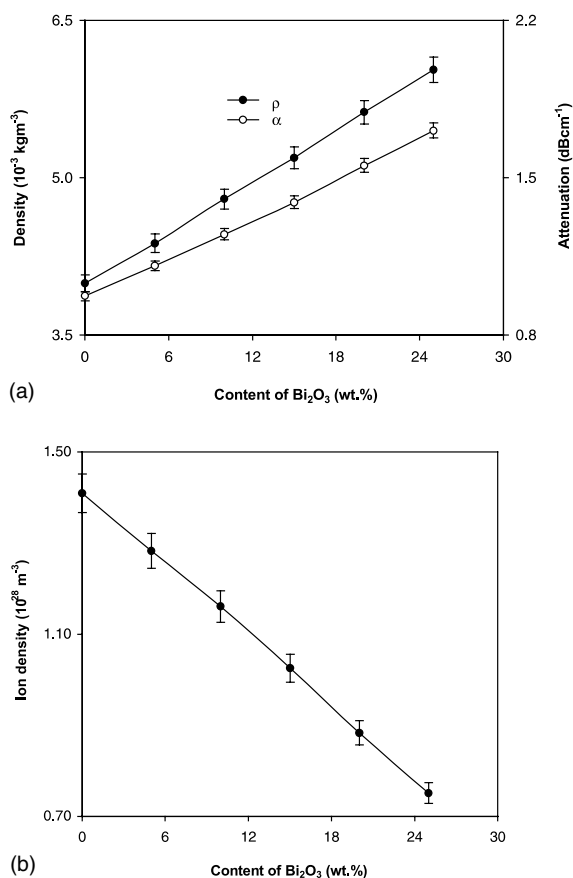


Fig. 5. Plot of (a) density and attenuation and (b) V-ion spacing with change in Bi_2O_3 content at room temperature. The lines are drawn for guides to the eye.

room temperature, ultrasonic velocities (Fig. 6(a)) show a linear decrease with addition of Bi_2O_3

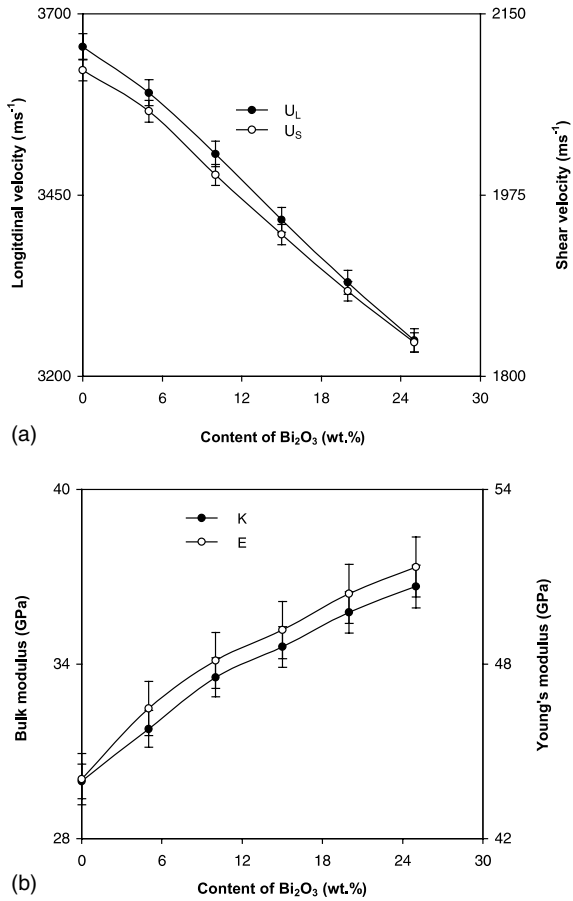


Fig. 6. Plot of (a) longitudinal and shear velocity and (b) bulk and Young's modulus with change in Bi_2O_3 content at room temperature. The lines are drawn for guides to the eye.

content without showing any anomalies as observed in many other TMO glasses. Elastic moduli show a monotonic increase with addition of Bi_2O_3 as shown in Fig. 6(b). (The longitudinal and shear modulus was not shown as its show a similar variation as that of Young's and bulk modulus.) Poisson's ratio obtained from elastic moduli almost remain constant, while Debye temperature obtained from ultrasonic velocities show a continuous decrease with addition of Bi_2O_3 as shown in Fig. 7(a). The dependence of bulk modulus with mean atomic volume is shown in Fig. 7(b).

The temperature dependence of longitudinal and shear velocity shows a gradual decrease (Fig. 8) in all Bi_2O_3 content without showing any

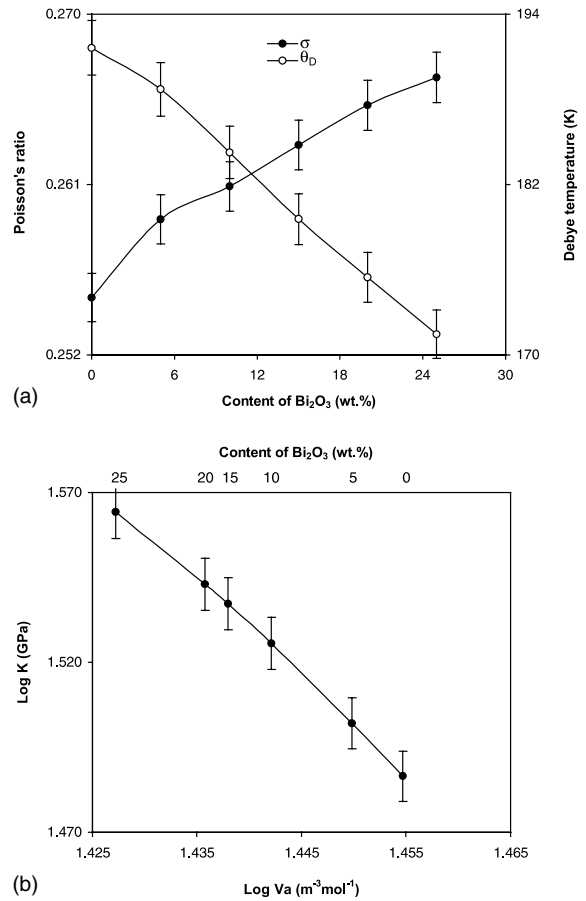
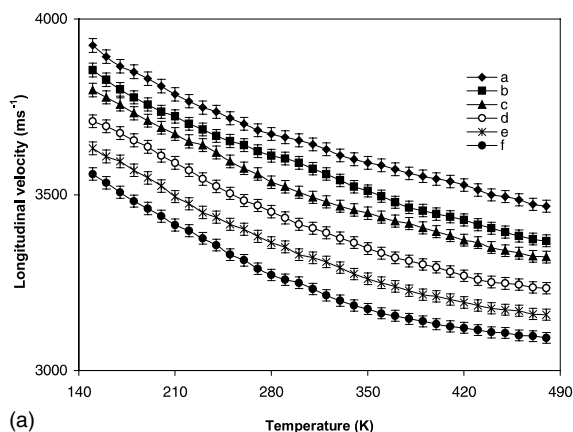
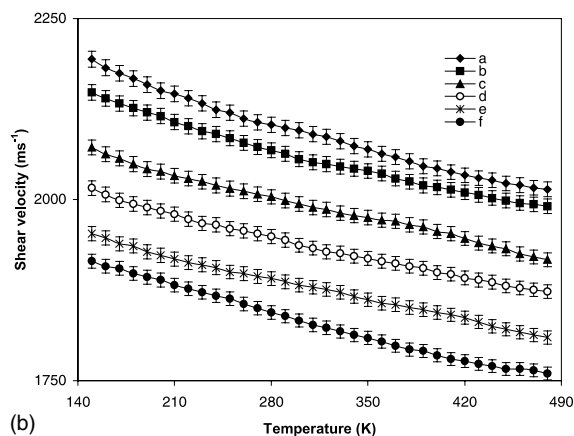


Fig. 7. Plot of (a) Poisson's ratio and Debye temperature with change in Bi_2O_3 content and (b) bulk modulus with change in molar mean atomic volume at room temperature. The lines are drawn for guides to the eye.

anomalies over the entire range of temperatures from 150 to 480 K. The temperature dependence of the relative percentage change in velocities is given in Fig. 9. The elastic moduli as a function of temperature (Fig. 10) show a similar trend of variation as that of velocities without showing any anomalies in the whole range of temperature studied. The temperature dependence of Debye temperature (Fig. 11) shows a similar trend of variation as that of moduli without showing any anomalies. The attenuation as a function of temperature shows a gradual increase with absence of anomalies with Bi_2O_3 content as represented in Fig. 12(a). The acoustic loss $Q^{-1} (= \alpha\lambda/\pi)$ obtained



(a)



(b)

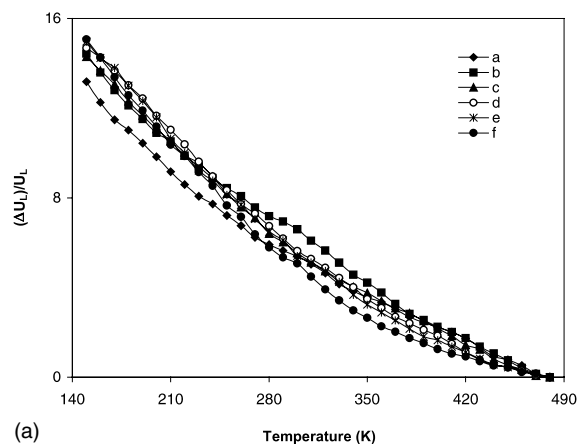
Fig. 8. Plot of (a) longitudinal velocity and (b) shear velocity with the temperature with change in Bi_2O_3 content. The lines are drawn for guides to the eye. (a) $50\text{V}_2\text{O}_5$ – 50TeO_2 (wt%); (b) $45\text{V}_2\text{O}_5$ – $5\text{Bi}_2\text{O}_3$ – 50TeO_2 (wt%); (c) $40\text{V}_2\text{O}_5$ – $10\text{Bi}_2\text{O}_3$ – 50TeO_2 (wt%); (d) $35\text{V}_2\text{O}_5$ – $15\text{Bi}_2\text{O}_3$ – 50TeO_2 (wt%); (e) $30\text{V}_2\text{O}_5$ – $20\text{Bi}_2\text{O}_3$ – 50TeO_2 (wt%); (f) $25\text{V}_2\text{O}_5$ – $25\text{Bi}_2\text{O}_3$ – 50TeO_2 (wt%).

from the attenuation (α) and the wavelength (λ) of the ultrasound increases with increase in temperature without showing any anomalies as shown in Fig. 12(b).

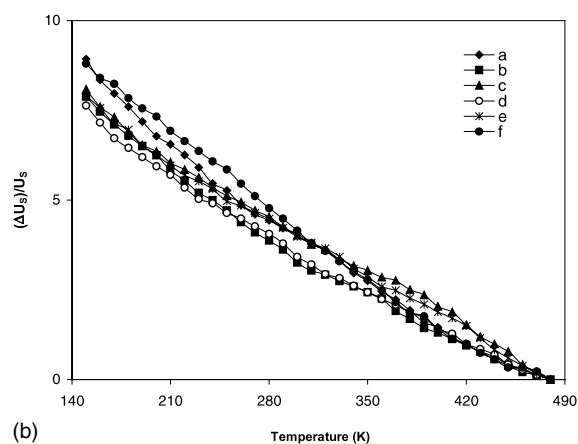
5. Discussion

5.1. Thermal behaviour

The glass transition temperature T_g is useful in exploring the difference in glass structure. The



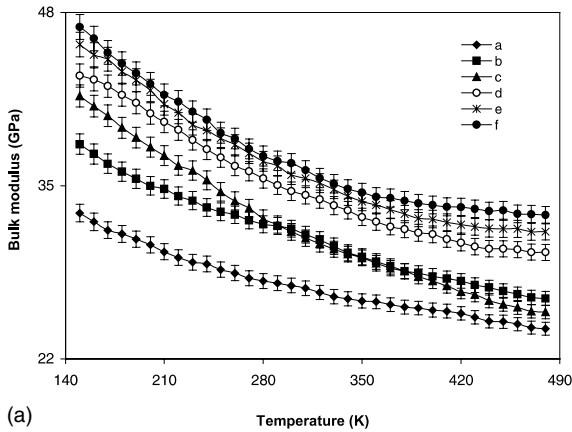
(a)



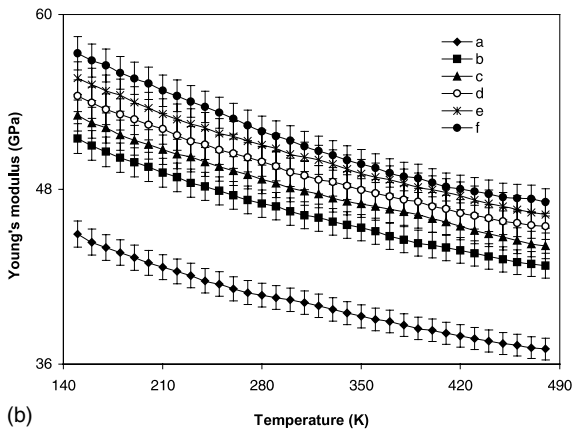
(b)

Fig. 9. Plot of relative percentage change in (a) longitudinal velocity and (b) shear velocity with change in temperature. The lines are drawn for guides to the eye. For key refer to Fig. 8.

thermal stability of the glass is a result of the glass structure. The thermally stable glasses will have close packed structure, while the unstable glasses will have loose packed structure [25]. In the case of open glass structure [1,25], the value of T_g declines. The values of T_g decreases from 690 to 220 °C with addition of V_2O_5 from 20–80 mol% of V_2O_5 in P_2O_5 containing vanadate glasses [26]. The T_g values (Table 1) in the present study increases from 230 to 396 °C with increase in Bi_2O_3 from 5 to 25 wt% (i.e. V_2O_5 decreases from 45 to 25 wt%). A similar upward trend of variation with increase in T_g with Bi_2O_3 [22] was noticed in the same glasses. Thus, in comparison, a low value of T_g in the present glasses with other glasses such as



(a)



(b)

Fig. 10. Plot of (a) bulk modulus and (b) Young's modulus with change in temperature. The lines are drawn for guides to the eye. For key refer to Fig. 8.

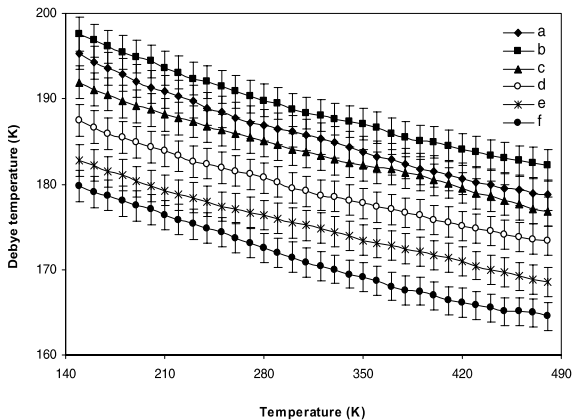
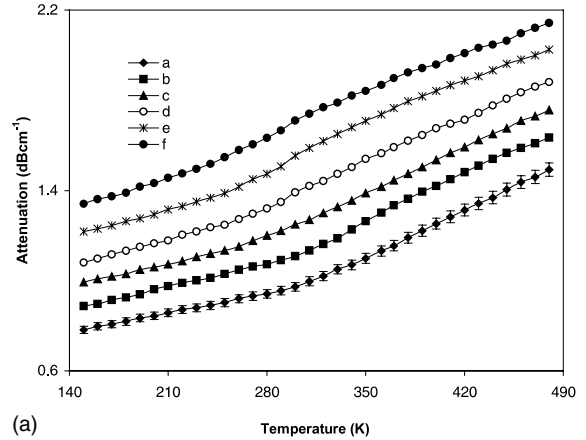
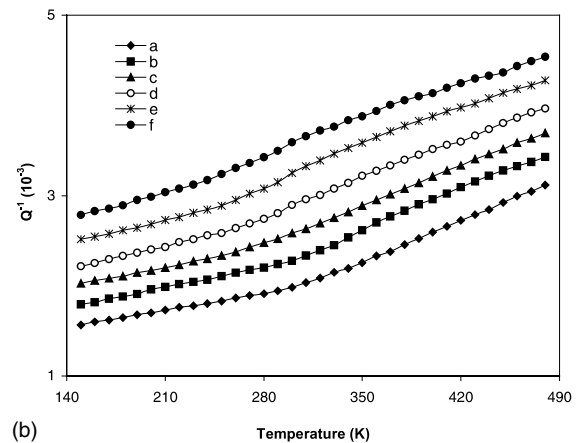


Fig. 11. Plot of Debye temperature with change in temperature. The lines are drawn for guides to the eye. For key refer to Fig. 8.



(a)



(b)

Fig. 12. Plot of (a) attenuation and (b) acoustic loss with change in temperature. The lines are drawn for guides to the eye. For key refer to Fig. 8.

$P_2O_5-V_2O_5$ [1,26] (T_g ranges from 690 to 220 °C for V_2O_5 from 20–80 mol%), $10F_2O_3-60V_2O_5-30P_2O_5$ ($T_g = 343$ °C) and $60V_2O_5-40P_2O_5$ ($T_g = 373$ °C) leads to the interpretation that vanadate bismuth tellurite glasses is thermally unstable in comparison with other glasses.

It is inferred from the Table 1 that the increase in Bi_2O_3 content (i.e. decrease in V_2O_5 content) causes a decrease in molar volume. On the other hand, the decrease in molar volume (i.e. increase in Bi_2O_3 content) results an increase in T_g (Fig. 4) indicating that the increase in Bi_2O_3 content contributes to the increase in thermal stability which

expands the glass formation region of this system as discussed in V_2O_5 – Bi_2O_3 – TeO_2 glasses [27].

5.2. Ultrasonics and elastic behaviour at room temperature

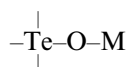
The addition of TMO to the highly conductive [8–10,22] tellurite glasses such as V_2O_5 – R – TeO_2 ($R = Sb_2O_3$ or Bi_2O_3 or ZnO or SnO) is quite interesting. When TMO is added to an oxide glass, in addition to the possible interaction between the alkali ions and oxygen atoms, structural rearrangement would also take place. As a result, the degree of structural compactness, modification of the geometrical configuration, etc. in the glass network is possible with change in composition. It is known that the structure of the tellurite glass is a laminar network [26] based on triangular TeO_3 pyramids or square TeO_4 pyramids with which a high TeO_2 content is open. However, in the present system, the content of TeO_2 is fixed as 50 wt% and we are introducing Bi_2O_3 as a third component in the ternary tellurite glass $(50 - x)V_2O_5$ – xBi_2O_3 – $50TeO_2$ at the expense of V_2O_5 . The replacement of V_2O_5 having low atomic weight (molecular weight of $V_2O_5 = 181.88$) by high atomic weight (molecular weight of $Bi_2O_3 = 465.96$) may leads to an increase in density (Fig. 5(a)) with increase in Bi_2O_3 content as we studied earlier [28].

The addition of small amount of Bi_2O_3 as a third component into the binary vanadium tellurite glasses resides in V_2O_5 – TeO_2 glass network as a modifier without entering into the glass matrix. A further increase in Bi_2O_3 content to the binary tellurite glasses, results in the formation of glass matrix. With more and more addition of Bi_2O_3 content, the following structural readjustment are presumed to take place: (i) The initial addition of Bi_2O_3 to the V_2O_5 – TeO_2 glasses, causes splitting of V – O – V and O – Te – O bonds and hence, the bridging oxygens (BOs) are converted into non-bridging oxygens (NBOs) and (ii) A further addition of Bi_2O_3 into glass interstices, more and more ions being open up in the network [27,29]. Thus, weakening of the glass structure or reduction in the rigidity of the network take place.

Generally, the formation of non-bridging oxygen is the most important influence of the alkali

oxide effect in turn produces a weakening (loose packing) of the glass network or reduction in rigidity of the glass. As is evident, a decrease in velocities (both longitudinal and shear) as observed (Fig. 6(a)) in the present study with addition of TMO. At this stage it is essential to consider the nature of Bi_2O_3 . It is an unconventional glass and its characteristic features are not explored well, seems to be difficult to understand the behaviour when Bi_2O_3 is added to glass such as vanadate–tellurite glasses.

The observed higher values in velocity (3855 and 2148 m s^{-1} respectively for longitudinal and shear waves) at low Bi_2O_3 (5 wt%) content and low values in velocity (3093 and 1760 m s^{-1} respectively for longitudinal and shear waves) at high Bi_2O_3 (25 wt%) content confirms a substantial change in glass structure. The observed reduction in velocities (Fig. 6(a)) explains the structural modification is associated with the modifier Bi_2O_3 role. Generally, the role of modifier Bi_2O_3 in tellurite glasses [30] contributes the following effects: (i) incorporation of holes and defects between two chains or layers; (ii) the formation of strong



ion covalent bonds; and (iii) strong influence on the $Te_{-ax}O_{eq}$ – Te angle between two polyhedra.

The structural change in the present glass is speculated by considering the transformation of TeO_4 tbp units to TeO_3 trigonal pyramid units [14,31]. It is evident from IR studies, the existence of Te as 3-coordinated in vanadium tellurite glasses [14] and both 3- and 4-coordinated structure in glasses at higher TeO_2 content. The existence of simultaneous formation of 3- and 4-coordinated tellurite structure has been studied in copper tellurite glasses [31] from IR spectra. In the present ternary tellurite glasses, both the 3- and 4-coordinated tellurite polyhedran are simultaneously exists (since TeO_2 is 50 wt%). In binary tellurite ($MO_{1/2}$ – TeO_2) glasses [32], the average co-ordination number of TeO_2 atom decreases from 4 to 3 upon incorporation of a network modifying oxide. Based on the above discussion, it is presumed that the introduction of Bi_2O_3 to the glass former TeO_2 leads to the formation of distorted TeO_{3+1} units

(where the subscript 3 + 1 shows the existence of a longer bond compared to the three others [33]) followed by a creation of regular TeO_3 trigonal pyramid units leaving non-bridging oxygen atoms. On the other hand, the formation of Te-O-Bi linkages is also possible due to the cross linkages between Bi_2O_3 and TeO_4 units. The formation of such Te-O-M linkages in addition to the NBO has been revealed on ternary silver tellurite glasses [13].

It is also inferred that the addition of modifier Bi_2O_3 in $\text{V}_2\text{O}_5\text{-Bi}_2\text{O}_3\text{-TeO}_2$ glasses break the random network leaving the NBO with the formation of clusters of point defects and also dislocations. The further addition of Bi_2O_3 causes more and more discontinuity [30] and, hence, a decrease in rigidity and velocity results. Similar observations have been made on tellurite glasses such as $\text{ZnF}_2\text{-PbO-TeO}_2$ [34] and $\text{Bi}_2\text{O}_3\text{-TeO}_2\text{-PbO}$ [19] glasses. Therefore, in the present glasses, a progressive transformation from a rigid frame work characteristics of TeO_2 glass former with addition of Bi_2O_3 modifier to a regular TeO_3 units and ionic character bonds (NBO) results in a decrease in rigidity and, hence, velocity (Fig. 6(a)). Normally, the decrease in rigidity of the glasses contributes a decrease in velocity and, hence, modulus. However, in the present work, an increase in modulus was noticed (Fig. 6(b)) with increase in Bi_2O_3 content. The decrease in network rigidity and sound velocity with addition of a modifier to the glass former TeO_2 due to the transformation of co-ordination ($4 \rightarrow 3 + 1 \rightarrow 3$) have been studied on tellurite glasses [33,35]. The observed above results on velocity variation in the present glass samples can also be compared with lead, since bismuth and Pb are heavy elements with nearly same atomic weight and their ions are isoelectronic. The behaviour of velocity, modulus, density and attenuation with addition of PbO (in the place of Bi_2O_3) in $\text{V}_2\text{O}_5\text{-PbO-TeO}_2$ [36] glasses exhibits the same as observed in $\text{V}_2\text{O}_5\text{-Bi}_2\text{O}_3\text{-TeO}_2$ glasses.

Attenuation of ultrasound waves (Fig. 5(a)) in binary vanadium tellurite glass is slightly less than that of Bi_2O_3 containing vanadium tellurite glasses where Bi_2O_3 is added into the network as a network modifier. In Bi_2O_3 containing ternary vana-

date glasses, a gradual increase in attenuation with addition of more and more Bi_2O_3 content has been noticed. This may be due to the following facts: the increase in Bi_2O_3 content in TeO_4 network results two types of NBO [21] i.e. Te-O_{eq} and Te-O_{ax} (where O_{eq} and O_{ax} refers to the oxygen in an equatorial plane and the oxygen in an axial position respectively) bonds as assumed in alkali tellurite glasses [32] and also due to the formation of point defects and dislocations as we discussed earlier. In binary copper oxide $(\text{CuO})_x\text{-(TeO}_2)_{1-x}$ glasses [21], a similar increase in attenuation with addition of CuO due to the breaking of bonds were studied.

The bulk modulus (K) and molar atomic volume (V_a) are related by $K V_a^m = C$ [37] (where C is a constant). The value of m for an oxide glass is 4 [37]. From the above results it is noted that a decrease in volume should lead to an increase in bulk modulus. In the present work, an increase in bulk modulus (Fig. 6(b)) with increase in Bi_2O_3 content was noticed. Thus, decrease in atomic volume leads to an increase in modulus (Fig. 7(b)). It is clear from the above results that the type of bonding in the network plays a dominant role in deciding the bulk modulus rather than volume. The observed increase in bulk modulus is associated with the change in cross linkage/co-ordination of the network [28]. Thus, the change in dimensionality of the glass enhances the rigidity of the network. The addition of Bi_2O_3 in VBT glasses increases the co-ordination number as 6 and known to exist as BiO_6 octahedral structural unit. The increase in Bi_2O_3 content increases the concentration of BiO_6 structural unit and hence, leads to an increase in moduli as observed (Fig. 6(b)). The existence of octahedral (BiO_6) structural unit and the dependence with Bi_2O_3 content has been revealed from EXAPS, X-ray, neutron scattering experiments and IR studies on bismuth based glasses [29,38–41].

Thus, the continuous addition of Bi_2O_3 content, the glass network becomes more discontinuous [30] and thus leads to an increase in glass density (Table 1) and lowers the Debye temperature (Fig. 7(a)), leading to an increase in modulus which is contradiction to the observed velocities (Fig. 6(a)). From these results, it is inferred that the structure

of these glasses is an independent combination of TeO_4 and Bi_2O_3 structures. The observed continuous increase in T_g (Table 1) and modulus (Fig. 6(b)) with increase in Bi_2O_3 content reveals the dominant role of BiO_6 octahedral units as represented in other Bi_2O_3 based glasses [27,29]. The addition of Bi_2O_3 leads to the formation of stable Te-O-M ion covalent bonds at the expense of bridging oxygen ions. The addition of Bi_2O_3 content with high TeO_2 (50%) content is sufficient to increase the density and, hence, modulus (Fig. 6(b)) as compared with the base $50\text{V}_2\text{O}_5$ – 50TeO_2 glass without modifying the structure. Even though the modulus is sensitive on structural changes in glasses as it depends on the change in cross link densities, the observation made on Poisson's ratio supports that there is no appreciable change in cross link densities with addition of Bi_2O_3 .

The change in cross link density of the glass network is well understandable from the variation in Poisson's ratio. In general, high cross link density has a Poisson's ratio of the order of 0.1 to 0.2 [18], while low cross link density has a Poisson's ratio between 0.3 and 0.5. In the present system, Poisson's ratio (Fig. 7(a)) is almost constant (changes from 0.25 to 0.27) when Bi_2O_3 content is increased from 5 to 25 wt% and this supports the above discussion. The observed Debye temperature, obtained from the ultrasonic velocity data, is particularly sensitive with Bi_2O_3 content (Fig. 7(a)). The observed decrease in θ_D with addition of Bi_2O_3 supports that the addition of Bi_2O_3 content indicates the loose packing structure of the glass structure with increasing the creation of NBOs as discussed above. Similar observations on increase in loose packing structure of the glasses with creation of more and more NBOs have been studied on TeO_2 – V_2O_5 – Ag_2O [42] and $\text{Bi}_{3.5}\text{PbO}_{0.5}\text{Sr}_3\text{Ca}_3\text{Cu}_4\text{O}_z + \text{Ag}_2\text{O}$ glasses [43].

The structural changes in V_2O_5 – TeO_2 – Bi_2O_3 glass are further supported from electrical conductivity measurements. In V_2O_5 – Bi_2O_3 – TeO_2 glasses, the contribution of Bi_2O_3 or TeO_2 to the mobility with change in adiabatic small polaron hopping (SPH) process and non-adiabatic SPH respectively when V_2O_5 is ≥ 50 mol% and <50 mol% is due to the hopping between V-Bi-V or V-

Te-V bonds [22]. The observed results suggest that the conductivity mainly depends on the cation radius of the third component (i.e. Bi_2O_3 or ZnO or SnO or Sb_2O_3). It was concluded that the addition of TMO as a third component results in highly conductive vanadium tellurite glasses and also a strong interaction between vanadium centre and glass network [7]. The present glass, Bi_2O_3 , is substituted interstitially in the place of V_2O_5 and, thus, V-O-V bonds are broken and new bridge bonds V-O-Bi and Bi-O-Bi are formed, as presumed in TeO_2 – V_2O_5 – Fe_2O_3 glasses [15]. The temperature and Bi_2O_3 composition dependence of conductivity show a decreasing trend in V_2O_5 – Bi_2O_3 – TeO_2 glasses [22], which is ascribed to the formation of the above bridge bonds.

5.3. Temperature effect of velocity and elastic moduli

The variation of velocities, elastic moduli, Poisson's ratio and attenuation as a function of temperature explore more information about the structural changes and stability of the glass network. The observed velocities (Fig. 8) and elastic moduli (Fig. 10) decreases monotonically without showing any anomalies with increase in temperature for different Bi_2O_3 contents, unlike many other glasses such as $(\text{CuO})_x$ – $(\text{TeO}_2)_{1-x}$ [21] and $(\text{Bi}_2\text{O}_3)_{1-x}$ – $(\text{CuO})_x$ [44]. Elastic moduli are more sensitive on structural changes; the observed monotonic decrease in moduli infers the absence of structural phase changes as observed in V_2O_5 – Bi_2O_3 – BaTiO_3 glasses [45]. The continuous decrease in moduli (Fig. 10) in all the glass samples (VBT0 to VBT25) with increase in temperature shows the loose packing structure of the glass network to take place giving rise to continuous reduction in rigidity and, hence, velocities (Fig. 8). The observed Debye temperature (Fig. 11) shows a similar trend of variation to that of moduli; even though it was obtained from ultrasonic velocity data, it supports the observation made in elastic moduli studies. The normal behaviour of the tellurite glasses i.e. softening of the acoustic modes with increasing in temperature has been marked in contrast to the anomalous behaviour of silica based glasses [46].

The maximum relative percentage change in velocities (Fig. 9) increases from 13.2% to 15.1% for longitudinal velocity with the change in Bi_2O_3 content, and the same were found to decrease with increase in temperature. It is also inferred that the glasses with low Bi_2O_3 content are more stable than with high Bi_2O_3 content. The early study [25] explains that the stability of the glass depends on glass structure i.e. the stable glass will have close packed structure, while the unstable glass will have loose packing. For a particular Bi_2O_3 content, say 5 wt%, the same were found to decrease from 13.2% to zero, and indicate the loose packing structure of the network. The validity of $\Delta U/U$ in explaining the loose packing/stiffening of the network structure has been used to identify the phase changes [28,45]. It is presumed that TeO_4 tbp units, which are present at room temperature, changed to TeO_3 trigonal pyramid units leading to non-bridging oxygen (NBO) as the temperature increases. A similar change in co-ordination of TeO_4 tbp to TeO_3 trigonal pyramid units with increase in temperature increasing the formation of NBOs has been revealed using Raman studies on $\text{Li}_2\text{O}-\text{TeO}_2$ glasses [47–49]. Thus, it is concluded from the above studies that the TeO_4 units are connected with weak structure and, hence, intermediate structures vary as temperature increases contributing to a continuous decrease in rigidity/moduli (Fig. 10) and, hence, velocities (Fig. 8) of the glass as observed. The variation in Poisson's ratio with temperature also supports the same observation as made in elastic moduli.

As we discussed earlier, the breaking of weak structure of TeO_4 creates more and more discontinuity/point defects in the glass network as temperature increases. Under these circumstances, larger amount of wave energies are required for the propagation of ultrasonic waves. This implies a continuous increase in attenuation (Fig. 12(a)) and acoustic loss (Fig. 12(b)) with increase in temperature in all the glass samples without showing any anomalies. The attenuation of longitudinal waves in chalcogenide glasses (As_2S_3 , $\text{As}_{20}\text{S}_{80}$, $\text{Ge}_{20}\text{S}_{80}$) shows a continuous increase as a function of temperature [50]. Further, in all the glass samples studied, a fast increase in attenuation towards the glass transition temperature was noticed as

noted in chalcogenide glass [50]. Thus, the observed results in attenuation and acoustic loss (Fig. 12(b)) generally suggest that the loose packing of the network structure due to the creation of more and more discontinuities and defects in the glasses.

6. Conclusions

In the present study, tellurite containing vanadium $(50-x)\text{V}_2\text{O}_5-x\text{Bi}_2\text{O}_3-50\text{TeO}_2$ oxide glasses for different Bi_2O_3 contents $x = 0, 5, 10, 15, 20$ and 25 wt% have been prepared by the rapid quenching method. The ultrasonic velocity both longitudinal and shear, attenuation, elastic moduli, Debye temperature and Poisson's ratio in all the glass samples have been measured as a function of temperature from 150 to 480 K. The amorphous nature of all the prepared glass samples was confirmed using XRD and SEM studies. EDAX studies confirms the chemical content of all the glass samples.

The following are the conclusions drawn in the present investigation:

1. For any fixed temperature, say at room temperature, the increase in Bi_2O_3 (modifier) content contributes a monotonic decrease in velocities (both longitudinal and shear), Debye temperature, mean ion density and an increase in attenuation and moduli without showing any anomalous behaviour.
2. The above result infers the structure of the glass network is an independent combination of TeO_4 and Bi_2O_3 structures. Thus, a continuous increase in the concentration of BiO_6 octahedral unit structure with addition of Bi_2O_3 , and a progressive transformation from a rigid formula character of TeO_2 with addition of Bi_2O_3 glass modifier to a matrix of regular TeO_3 units and ionic behaviour bonds (NBOs), results in loose packing of the glass network.
3. A continuous decrease in velocities, elastic moduli and an increase in attenuation and acoustic loss as a function of temperature in all the glass samples exhibits without showing any anomalies unlike many other TMO glasses.

4. It is inferred from the above studies that the loose packing structure of the network is due to the creation of more and more discontinuities and defects in the glasses due to the breaking of weak structure of TeO_4 as temperature increases.
5. The relative percentage change in velocity further supports that the glasses with low Bi_2O_3 (close packed) content are more stable than with high Bi_2O_3 content (loose packed structure).

The present study confirms that the ultrasonic velocities, attenuation, elastic moduli and other related parameters are more sensitive in exploring the structure, stability and mechanical properties of vanadium tellurite glasses as a function of temperature.

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